

Run 05 | Fear of failure

Global (including India)

IV + controls + NES

Target Fear of failure prevents starting a business; binary 1 vs. 0
Sample 1,334,667 respondents **Coverage** 89 countries; 2015–2022
Positive cases 583,346 (43.71%) **Case time** 39m 45s
Train / val. / test 907,575 / 160,159 / 266,933 (68% / 12% / 20%, stratified)

SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

NES controls (12): Financing, Government support, Taxes and bureaucracy, Government programs, School entrepreneurship education, Higher/vocational entrepreneurship education, R&D transfer, Commercial/legal/professional services, Market changes, Market entry, Physical infrastructure, Cultural and social norms. Total predictors: 47.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.6976). Highest F1: Random Forest (0.6444). For Random Forest, adding NES changed ROC-AUC by -0.0011 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.6976	0.6346	57.74	50.96	87.63	0.6444	7m 10s
XGBoost	0.6815	0.6109	57.20	50.61	85.92	0.6370	23s
H2O GBM	0.6802	0.6092	56.78	50.32	86.60	0.6366	2m 55s
CatBoost	0.6773	0.6066	57.00	50.48	85.65	0.6352	2m 38s
Logistic Regression	0.6274	0.5501	50.03	46.40	92.46	0.6179	1m 11s
Lasso	0.6273	0.5503	50.77	46.75	91.03	0.6178	4m 53s
Elastic Net	0.6273	0.5503	50.77	46.75	91.03	0.6178	4m 58s
Ridge	0.6271	0.5501	50.00	46.39	92.45	0.6178	8m 45s
Decision Tree	0.6120	0.5297	43.79	43.74	99.88	0.6083	38s
Naive Bayes	0.5891	0.5214	44.07	43.83	99.27	0.6081	5m 21s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 6

Rank	Predictor	Raw importance	Share%
1	Age	15,585.83	11.23
2	Household size	8,619.12	6.21
3	Occupational status	8,491.15	6.12
4	Education attainment	6,342.02	4.57
5	Household income	6,007.51	4.33
6	Self-efficacy	5,390.08	3.88

Share is the percentage of total forest impurity importance across all 88 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	1,869.80	1.35
2	Economic quality	1,739.65	1.25

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Leading NES predictors within the same forest

Rank	NES predictor	Raw importance	Share%
1	Commercial/legal/professional services	1,289.60	0.93
2	Market changes	1,284.31	0.93
3	Cultural and social norms	1,223.15	0.88

NES shares use total forest importance.

Source: Completed Objective 4 NES extension batch, 29 September 2026; 01_obj4_global_iv_cv_nes_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.